

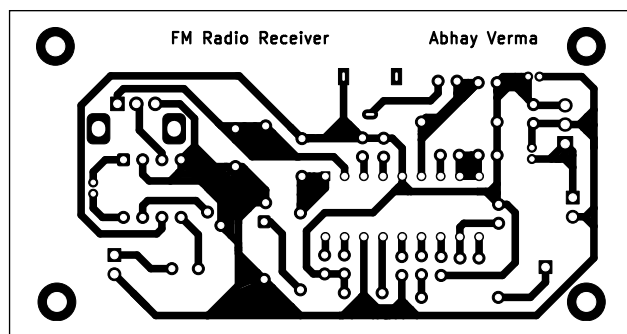


Fig. 3: Actual-size PCB

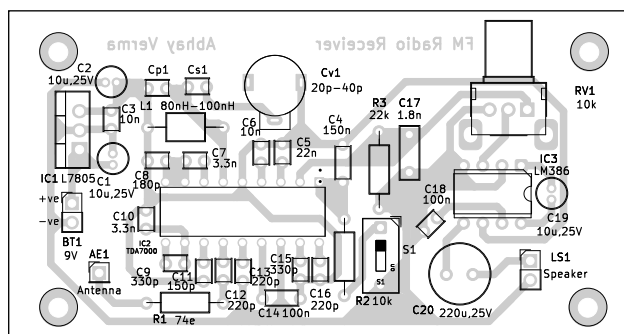


Fig. 4: Component layout

Construction and testing

An actual-size, single-sided PCB layout for the FM receiver is shown in Fig. 3, and its component layout in Fig. 4. After assembling the circuit on the PCB, mount it inside a suitable enclosure. Provide an opening for the variable capacitor CV1 and potentiometer RV1 so that the tuning and volume controls are easily accessible from the front or side panel of the enclosure.

Place the 9V battery inside the

enclosure and secure it properly.

Also, provide suitable holes or a grille in front of the loudspeaker to allow clear sound output without obstruction. Once assembled, switch on the receiver and adjust CV1 to tune to the desired FM station, while RV1 can be used to set the required volume level.

Alternatively, the circuit can be assembled on a breadboard for easy prototyping, testing, and modification, as shown in Fig. 1. While using a breadboard, ensure that all wiring is kept as short as possible to minimise noise pickup and signal interference. The antenna wire should be properly extended, and the prototype should preferably be tested in an open area for improved FM signal reception.

Decoupling capacitors should be placed close to the IC pins to suppress unwanted noise and enhance circuit stability. In addition, a proper common ground connection should be maintained throughout the circuit to ensure reliable operation and consistent performance.

For testing, connect a 9V battery (BT1) and verify that the L7805 regulator provides a stable 5V output to the TDA7000 FM receiver and LM386 audio amplifier. Connect an

FM antenna and slowly adjust the variable capacitor CV1 to tune across the 88MHz–108MHz FM broadcast band. Audio volume can be adjusted using the 10kΩ potentiometer (RV1). The tuning coil (L1) consists of four turns of 22 SWG copper wire wound on a 5mm-diameter former.

If reception is weak, check the antenna connection and slightly adjust the turns of L1 to fine-tune. The 10μF capacitor connected between pins 1 and 8 of the LM386 increases the amplifier gain, while the 100nF capacitor improves stability and noise filtering. When the circuit is properly assembled and tuned, the receiver provides clear FM reception with good sensitivity.

Learning points

This system demonstrates how FM signals are received, tuned, demodulated, and converted into audible sound using the TDA7000 FM receiver IC. It highlights the importance of stable power regulation with the L7805 and shows how the LM386 audio amplifier boosts the weak demodulated signal to drive a speaker.

The system also provides practical experience in RF circuit design, frequency tuning, audio amplification, noise reduction, grounding, and antenna connections. Overall, it offers a solid introduction to RF electronics and analogue signal processing, illustrating how different stages work together to form a complete FM radio receiver system. **EFY**

Abhay Verma is an electronics engineer at EFY and a passionate electronics hobbyist with a keen interest in practical circuit design and DIY projects.

PARTS LIST

Semiconductors:

- IC1 - L7805, 5V voltage regulator
- IC2 - TDA7000 FM receiver IC
- IC3 - LM386 audio amplifier

Resistors (all 1/4-watt, ±5% carbon):

- R1 - 74Ω
- R2 - 10kΩ
- R3 - 22kΩ
- RV1 - 10kΩ potmeter

Capacitors:

- C1, C2, C19 - 10μF, 25V electrolytic capacitor
- C3, C6 - 10nF ceramic disc capacitor
- C4 - 150nF film capacitor
- C5 - 22nF ceramic disc capacitor
- C7, C10 - 3.3nF ceramic disc capacitor
- C8 - 180pF ceramic disc capacitor
- C9, C15 - 330pF ceramic disc capacitor
- C11 - 150pF ceramic disc capacitor
- C12, C13, C16 - 220pF ceramic disc capacitor
- C14, C18 - 100nF ceramic disc capacitor
- C17 - 1.8nF film capacitor
- C20 - 220μF, 25V electrolytic capacitor
- Cv1 - Trimmer capacitor (20pF–40pF)
- Cs1 - 2pF to 22pF ceramic disc capacitor
- Cp1 - 22pF to 40pF ceramic disc capacitor

Inductors:

- L1 - 4 turn, 22 SWG, 5mm diameter (80nH–100nH)

Miscellaneous:

- AE1 - Antenna
- BT1 - 9V battery
- S1 - On/off switch
- LS1 - 8Ω, 0.5W speaker
- Designed PCB/breadboard
- Connecting/jumper wires
- Enclosure (optional)

TO DOWNLOAD PCB AND RESOURCES



Scan this QR code or go to <https://sourcecode.electronicsforu.com> to access and download PCB and other resources for this project